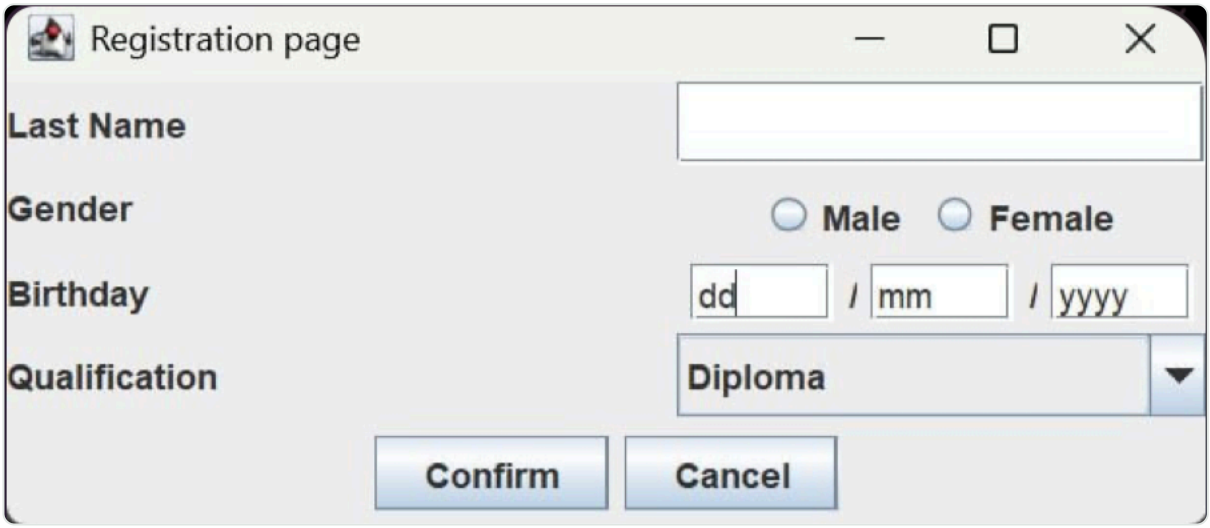
CS342 · VISUAL PROGRAMMING

Midterm 472 · Window Solution

سؤال النافذة · محارب · Registration page

GUI · Registration page Problem

Complete the class Registration so that it displays exactly the window below. The class header, the constructor header and super(title) are given.




```
public class Registration extends JFrame {

    public Registration(String title) {
        super(title);

        // ..... complete .....

    }
}
```

01 From the picture to the code

في الصورة	في الكود
كل سطر: كلمة يسار وشي يمين، 4 سطور	GridLayout(4, 2) في الـ CENTER
Male و Female في خانة وحدة	JPanel صغير فيه 2 JRadioButton
dd / mm / yyyy في خانة وحدة	JPanel صغير فيه 3 JTextField و 2 JLabel("/")
Diploma مع سهم	JComboBox فيه 4 خيارات
الأزرار تحت بالنص لحالها	JPanel في الـ SOUTH

```
BorderLayout
├── CENTER → GridLayout(4, 2)
└── SOUTH → [Confirm] [Cancel]
```

ليش مو GridLayout(5, 2) بس؟ لأن الأزرار في الصورة بالنص تحت الاثنين، مو في خانة وحدة. فتحطها في SOUTH لحالها.

02 The solution

```
import javax.swing.*;
import java.awt.*;

public class Registration extends JFrame {
    private JTextField tName, tDay, tMonth, tYear;
    private JRadioButton rMale, rFemale;
    private JComboBox<String> cQual;
    private JButton bConfirm, bCancel;

    public Registration(String title) {
        super(title);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        JPanel panel = (JPanel) getContentPane();
        panel.setLayout(new BorderLayout());

        JPanel center = new JPanel(new GridLayout(4, 2));     // 4 rows, label | component

        tName = new JTextField(15);
        center.add(new JLabel("Last Name"));
        center.add(tName);

        rMale = new JRadioButton("Male");
        rFemale = new JRadioButton("Female");
        JPanel pGender = new JPanel();
        pGender.add(rMale);
        pGender.add(rFemale);
        center.add(new JLabel("Gender"));
        center.add(pGender);
```

```
        tDay = new JTextField("dd", 4);
        tMonth = new JTextField("mm", 4);
        tYear = new JTextField("yyyy", 4);
        JPanel pBirth = new JPanel();
        pBirth.add(tDay);   pBirth.add(new JLabel("/"));
        pBirth.add(tMonth); pBirth.add(new JLabel("/"));
        pBirth.add(tYear);
        center.add(new JLabel("Birthday"));
        center.add(pBirth);

        String[] q = { "Diploma", "Bachelor", "Master", "PhD" };
        cQual = new JComboBox<String>(q);
        center.add(new JLabel("Qualification"));
        center.add(cQual);

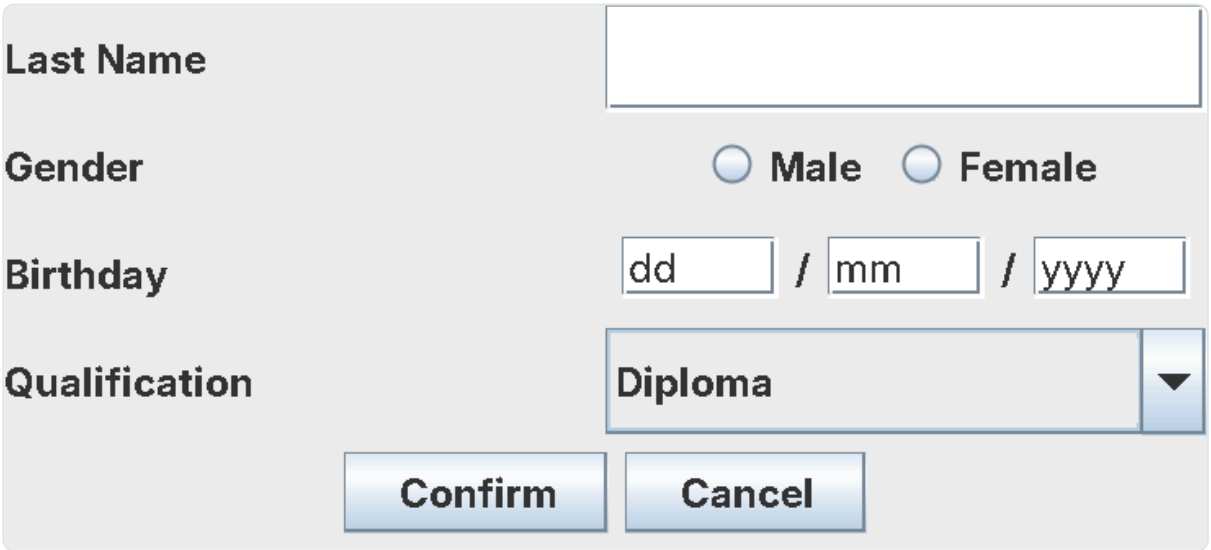
        panel.add(center, BorderLayout.CENTER);

        bConfirm = new JButton("Confirm");
        bCancel = new JButton("Cancel");
        JPanel south = new JPanel();                          // FlowLayout: centered
        south.add(bConfirm);
        south.add(bCancel);
        panel.add(south, BorderLayout.SOUTH);

        this.pack();
        this.setVisible(true);
    }

    public static void main(String[] args) {
        new Registration("Registration page");
    }
}
```

The result 03



لو تبي Male و Female ما ينختارون مع بعض

تضيف `ButtonGroup g = new ButtonGroup(); g.add(rMale); g.add(rFemale);`.